SE LAB1

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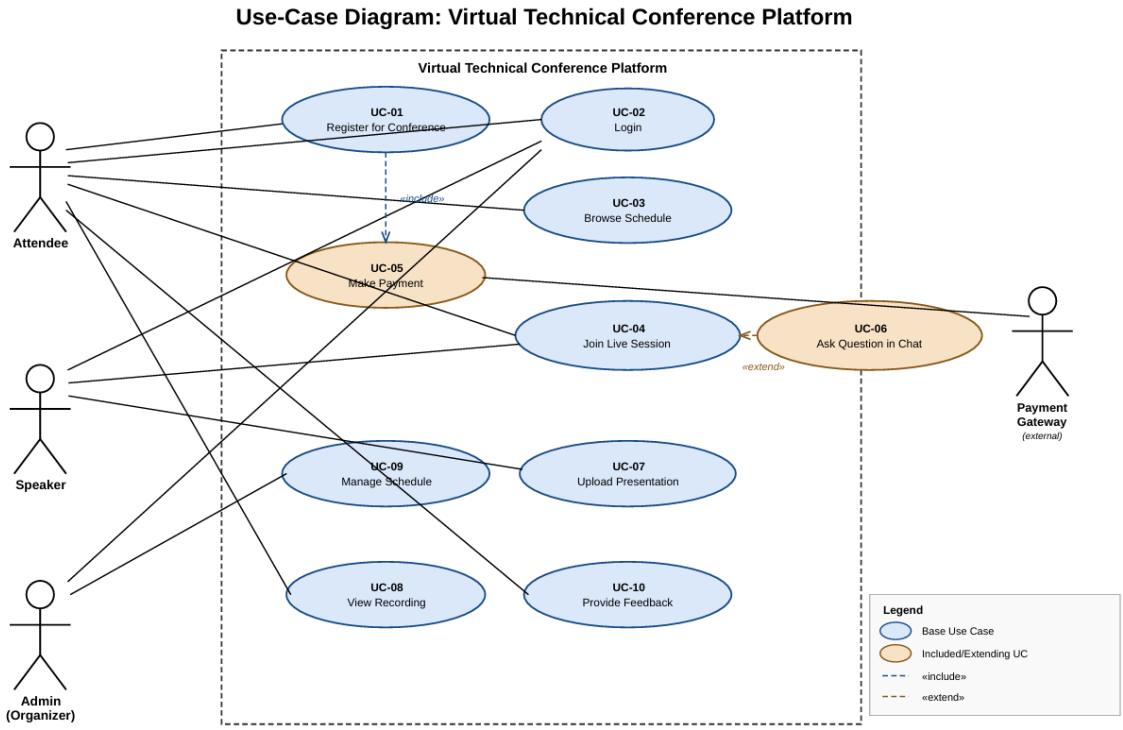
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SECTION-B

1. Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow an attendee to register for the conference by creating an account with a valid email address, password, and selecting a ticket type (Free / Standard / Premium).	High	When an attendee submits a valid email, password, and ticket type, a confirmation email is sent within 60 seconds and the account status changes to "Registered."	Registration is the entry point for all attendee interaction with the platform.
FR-002	Functional	The system shall allow an attendee to browse and filter the session schedule by track, date, and speaker name.	High	When an attendee applies a filter (e.g., track = "AI/ML"), only matching sessions are displayed within 2 seconds.	Attendees need to quickly locate relevant sessions across a multi-track program.
FR-003	Functional	The system shall allow an attendee to join a live streaming session by clicking "Join Now" no earlier than 10 minutes before the scheduled start time.	High	Clicking "Join Now" within the allowed window opens the live video stream within 5 seconds; clicking earlier displays a countdown instead.	Core functionality that delivers the conference content to remote attendees.
FR-004	Functional	The system shall allow a speaker to upload presentation slides (PDF/PPTX) and a pre-recorded backup video for their assigned session at least 24 hours before the session start time.	Medium	A file uploaded in an accepted format (PDF, PPTX, MP4) shows status "Upload Successful" and is retrievable from the speaker's session dashboard.	Backup materials ensure sessions can proceed if a speaker's live connection fails.
FR-005	Functional	The system shall allow an attendee to submit text questions to the speaker through a live chat panel during a session.	Medium	A submitted question of up to 280 characters appears in the moderator queue within 3 seconds of submission.	Live Q&A increases attendee engagement and mirrors in-person conference interaction.
NFR-001	Nonfunc.	The system shall support at least 5,000 concurrent live-stream viewers per session without the video quality dropping below 720p.	High	A load test with 5,000 simulated concurrent viewers shows average video resolution ≥ 720p and buffering time < 2 seconds.	The platform must reliably scale to conference-sized audiences without degrading the viewing experience.
NFR-002	Nonfunc.	The system shall encrypt all payment transactions in transit using TLS 1.2 or higher and shall not store raw card details on platform servers.	High	A security scan confirms all payment endpoints negotiate TLS 1.2+ only, and a database audit confirms no raw card numbers are stored.	Protects attendee financial data and ensures compliance with PCI-DSS requirements.

2. UML Use-Case Diagram



3.Use case Flow

Use Case: Join Live Session (includes UC-05 Make Payment via UC-01 Register)

Primary Actor: Attendee

Secondary Actors: Speaker (session host), Payment Gateway (external system, invoked earlier during registration)

Preconditions: The attendee has an active account (UC-02 Login) and a confirmed registration with payment completed (UC-01 → «include» → UC-05).

Trigger: The attendee wants to attend a live session listed on their conference schedule.

Main Success Scenario

1. Attendee logs in and navigates to "My Schedule."
2. System displays the list of sessions the attendee is registered for, with a "Join Now" button next to each.
3. Attendee clicks "Join Now" for a session that is starting within 10 minutes.
4. System verifies the attendee's registration and session timing.
5. System opens the live video stream and chat panel in the attendee's browser.
6. Speaker's live video and audio begin playing for the attendee.

7. Attendee watches the session and may optionally use UC-06 «extend» Ask Question in Chat.
8. System logs attendance (timestamp and duration) for the certificate/report.
9. Session ends; system displays a "Session Ended" screen with a link to UC-10 Provide Feedback and UC-08 View Recording.
10. Use case ends successfully.

Alternate Flow

4a. Session Not Yet Open / Registration Not Found

- 4a1. System displays "This session is not open yet" with a live countdown timer, or "Registration required" if the attendee is not registered for that track.
- 4a2. If registration is required, the attendee is redirected to UC-01 Register for Conference.
- 4a3. Attendee returns to "My Schedule" once the countdown ends or registration is complete, and re-attempts step 3.

5a. Stream Fails to Load

- 5a1. System displays a "Connection Error" message and automatically retries the stream connection up to 3 times.
- 5a2. If all retries fail, the system offers the speaker's pre-uploaded backup video (from UC-07) as a fallback.
- 5a3. If the attendee declines the fallback, the use case ends in failure and the attendance log records "Not Attended – Technical Issue."

Postconditions: Attendance is recorded for the attendee, and the session is available for feedback and, if recorded, for later viewing.